

# JAY ZHAO

zhaoj06@bu.edu | (518) 812-5682 | www.linkedin.com/in/jzhaonys | Boston, MA

## EDUCATION

### Boston University, College of Engineering

Boston, MA

Bachelor of Science, Mechanical Engineering

Expected May 2028

- Concentration in Aerospace Engineering
- Relevant Coursework: Engineering Design, CAD and Machine Components, Engineering Mechanics, Programming
- Activities & Societies: Terrier Motorsports (Formula SAE), Sigma Chi Fraternity, Music Business Club, EK100 Freshmen Student Advisor

## EXPERIENCE

### Engineering Intern

Jun 2026 - Aug 2026

Becton Dickinson (BD)

Glens Falls, NY

- Supported multiple Class III Medical Device product lines by executing cost saving and quality driven projects
- Utilized SolidWorks and 3D printing to re-design mounting rack gear system, reducing repair frequency and material cost savings by 90%, saving \$135,000 annually
- Evaluated feasibility of expanding acceptance criteria for design specification and presented findings for implementation, resulting in immediate savings of \$50,000 and \$15,000 in scrap reduction savings annually
- Designed and implemented new cutting fixture for Ureteral Catheters line, increasing processing efficiency by 25%
- Participated in 1 external and 2 internal manufacturing Kaizen events, by collecting and interpreting cycle time and product output trends. Performed feasibility testing and investigated process variation, supporting a total annual savings of \$200,000

### Wait Staff

May 2025 - May 2026

Thai Sushi Aroy Dee

Glens Falls, NY

- Processed 75+ customer orders per night using a digital POS system, executed opening and closing procedures, including setup, sanitation, and equipment security, to maintain efficient service and safe operations

## PROJECTS

### Ergonomic Steering Wheel

May 2026 - Present

- Currently own the design, development and manufacturing of the '26-'27 steering wheel for the Terrier Motorsports Formula SAE car, CAD modeling assembly using Autodesk Fusion and redesigning grip geometry with an emphasis on ergonomics and driver comfort
- Integrated on-wheel electronics and designed components for manufacturability using ABS 3D printing and aluminum
- Performed basic FEA to evaluate stresses and validate the steering wheel's structural integrity under expected loading

### Automatic Color Sorter

Feb 2026 - Apr 2026

- Collaborated with 3 peers to design and develop an Arduino-based color sorting system capable of identifying and separating colored candies at an accuracy rate >90%, and under a budget of \$200
- Modeled a modular housing system in SolidWorks, manufacturing parts via 3D-printing and laser cutting MDF, assembled and wired electronic systems
- Developed and iteratively refined test code to calibrate color sensor and motor, improving actuation accuracy while preventing candies from escaping system, resulting in a color identification rate of 100% an accuracy rate of 92% and a total cost of \$56

### Room Temperature Monitor Box

Feb 2025 - Apr 2025

- Created an Arduino-based ambient temperature monitor using a TMP36 sensor and LCD to display real-time data, with a PKM17 buzzer for alerts above 90°F
- Modeled a custom 9V battery holder and enclosure in Onshape and conducted CAD-based assembly of components to validate fit, tolerances, and system integration before 3D-printing parts
- Programmed an algorithm to acquire and process TMP36 sensor data, testing and refining the code in order to improve measurement accuracy and overall system performance

## SKILLS

**Software:** SolidWorks, OnShape, Autodesk Fusion, MATLAB, Python, Cura, Microsoft Office, Google Suite

**Technical:** CAD Modeling, Soldering, 3D-printing, Wiring/Circuit Assembly, Laser Cutting, Manual Mill/Lathe